



EENG 385 - Electronic Devices and Circuits
Frequency Domain: Active Filters
How To: Use Simulator to Build a Bode Plot Using Frequency Sweep Method

How To: Simulation Bode Plot – Frequency Sweep

In the previous How To, you built a Bode plot for the circuit shown in Figure 1.

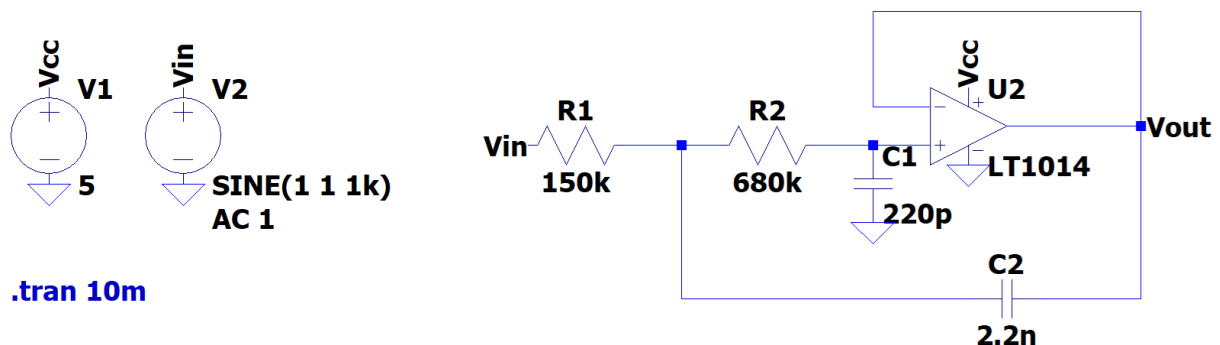


Figure 1: The low-pass filter on the Audio board is built around an op amp powered by 5V and gnd.

Now that you have gone through the painful process of building a Bode plot by hand, you will be delighted to know there is a far easier method to do this, have LTspice do it for you; the AC Analysis tool. The AC Analysis function runs a series of simulations, each:

- Will have an incrementally higher frequency for the AC source, V2 in Figure 1. You set the range of frequencies for the AC source,
- Will record the amplitude of each probe in the circuit in decibels with respect to the amplitude of the AC source,
- Will record the phase of each probe in the circuit in degrees with the phase of the AC source being 0°,

The waveform window will display the recorded data as magnitude and phase versus frequency. The frequency axis is logarithmic, forming a Bode plot for each probe in the circuit.

Let's try this out for our low-pass filter in Figure 1. Start by selecting the Configuration Analysis



icon in the top toolbar. In the pop-up, select the AC Analysis tab. We will use the same settings as you used in the point-by-point method to generate the Bode plots in Excel.

- Type of sweep: Decade
- Number of points per decade: 20
- Start frequency 1Hz
- Stop frequency 47kHz

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Click OK and your cursor is replaced with the text “.ac dec 6 1 47k”. Place this text near your AC source then run the simulation by clicking the green arrow in the top toolbar. The output shown in Figure 2 has the attenuation scale on the left axis and the phase scale on the right axis.

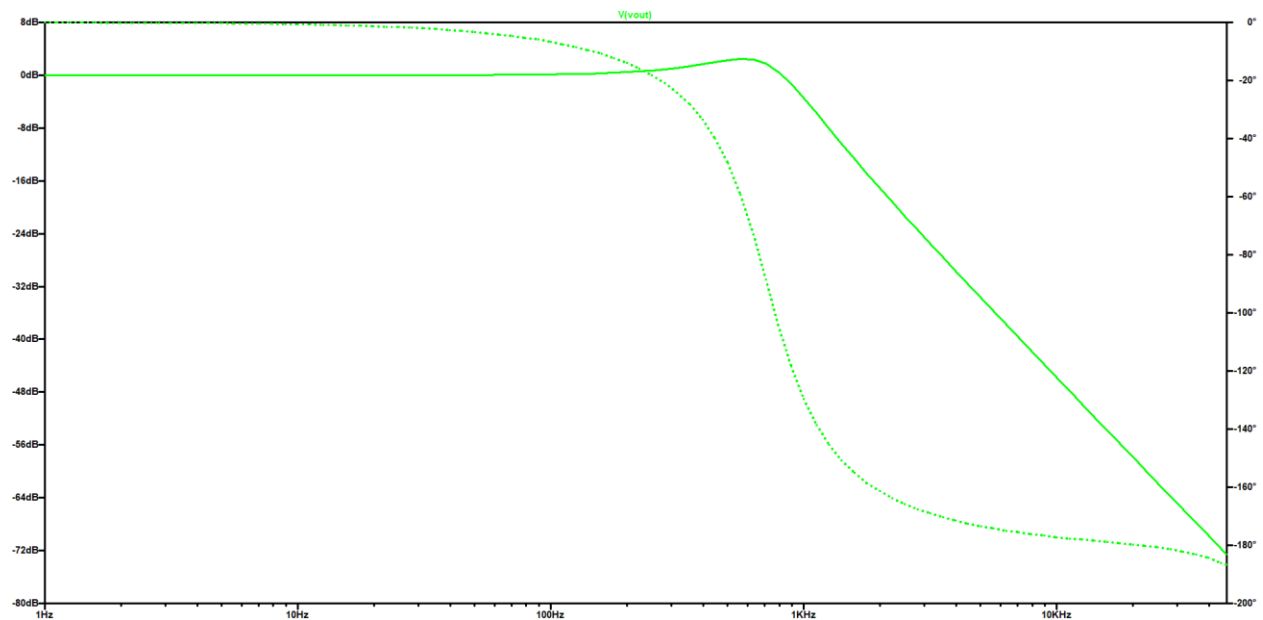


Figure 2: The grapher output after running a frequency sweep on the circuit in .

You are now ready to generate all the Bode plots needed for this lab, good luck!